

2024 Another Big Year for Space Exploration

As we enter the new year, we take a look back at some of the biggest space stories of 2024.

Historic private moon landing

Nearly one year ago, a private spacecraft completed the first U.S. moon landing in more than 50 years. The American space agency NASA announced the space lander, called Odysseus, had touched down on the lunar surface February 22.

Odysseus was developed by the Texas-based company Intuitive Machines. The spacecraft carried out several science experiments before powering down permanently on the moon in late March.

NASA said Odysseus had successfully captured data to help the agency “better understand the moon’s environment and improve landing precision and safety.”

Japan Moon Landing

Japanese moon landing

Japan’s space agency JAXA announced in January 2024 that its SLIM spacecraft had successfully landed on the moon. But the lander touched down the wrong way up, causing some communication and power problems.

Japanese space officials reported that they stayed in communication with SLIM through late April. They said the spacecraft was able to collect valuable data about the touchdown and surrounding area.

One purpose of the mission was to demonstrate new precision landing technology. JAXA said the experiment was successful because SLIM was able to land within 10 meters of its planned target.

In this photo released by China's Xinhua News Agency, the the unmanned Chang’e 5-T1 spacecraft is launched atop an advanced Long March 3C rocket from the Xichang Satellite Launch Center in southwest China's Sichuan Province, Friday, Oct. 24, 2014. (AP Pho

Chinese moon landing and return

In May, China launched its Chang’e 6 spacecraft to the moon’s far side. In late June, Chinese space officials announced the spacecraft had returned to Earth with rock and soil samples collected on the lunar surface.

The successful return made China the first country to ever collect and bring back lunar samples from the far side of the moon. The area is known as the far side because it always faces away from Earth.

The European Space Agency plans to launch its Hera spacecraft in October on a mission to return to an asteroid system visited by NASA's DART spacecraft in 2022. (ESA)

Europe's asteroid exploration mission

In October, the European Space Agency (ESA) said it had successfully launched a spacecraft, called Hera, to explore an asteroid struck in an earlier NASA test mission. Hera's trip to the asteroid is expected to take about two years.

Hera is headed to the asteroid Dimorphos. NASA crashed its DART spacecraft into the space rock in 2022, about 11 million kilometers from Earth. NASA said its mission successfully showed it was possible to change the orbit of asteroids that might threaten Earth in the future.

Hera aims to closely examine Dimorphos to get more details on how the strike by DART affected the asteroid. The data that NASA collected is to be used together with information gathered by ESA to develop future planetary defense plans.

This artist's concept depicts NASA's Europa Clipper spacecraft in orbit around Jupiter. (Image Credit: NASA/JPL-Caltech)

Europa Clipper to explore Jupiter's moon

NASA announced the successful launch of its Europa Clipper spacecraft in October. The explorer is designed to search for possible signs of life on Jupiter's moon Europa.

Jupiter is the largest planet in our solar system and has numerous moons. But icy Europa has been identified by scientists as a good candidate for having the right conditions to support life. Several studies have suggested the moon likely contains a large ocean beneath its icy surface.

NASA said the Europa Clipper would fly past Mars, then back toward Earth, "using the gravity of each planet to increase its momentum." These kinds of "gravity assists" are expected to provide the needed push to get the spacecraft to Jupiter by 2030.

In this photo provided by NASA, Boeing Crew Flight Test astronauts Butch Wilmore, left, and Suni Williams pose inside the International Space Station's Harmony module and Boeing's Starliner spacecraft on June 13, 2024. (NASA via AP)

Boeing's crew flight to the ISS

In early June, Boeing successfully carried out a test mission for NASA by transporting two American astronauts – Butch Wilmore and Suni Williams – to the International Space Station (ISS).

The two were expected to stay in space for only a week. But ongoing technical issues with the Starliner spacecraft prevented a return trip.

NASA announced plans in August to have Starliner return to Earth without the astronauts. The current plan is for Wilmore and Williams to return on a SpaceX Dragon spacecraft sometime in late March or April.

This image provided by SpaceX shows SpaceX's mega rocket booster returning to the launch pad to be captured during a test flight Sunday, Oct. 13, 2024, in Boca Chica, Texas. (SpaceX via AP)

SpaceX landing demonstration

SpaceX has now completed six launch tests of its huge Starship rocket system. Four of those tests were carried out in 2024. The last test took place in November.

The Starship rocket and spacecraft system is designed to be a completely reusable vehicle to transport astronauts and supplies. It has successfully demonstrated that ability in its most recent experimental flights.

During one of the tests in October, Starship's Super Heavy booster rocket was "caught" and secured by a huge structure that includes metal arms. A short time later, Starship's second stage safely splashed down in the Indian Ocean.

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English.

Quiz - 2024 Another Big Year for Space Exploration

Vocabulary

precision - n. ความแม่นยำ, ความถูกต้องแน่นอน

sample - n. ตัวอย่าง (สำหรับทดสอบหรือศึกษาทางวิทยาศาสตร์)

asteroid - n. ดาวเคราะห์น้อย

mission - n. (การสำรวจอวกาศ) การเดินทางของยานอวกาศไปยังเป้าหมายพร้อมกับภารกิจที่คาดว่าจะดำเนินการ

momentum - n. แรงส่ง, แรงเฉื่อยที่ทำให้สิ่งต่างๆ เคลื่อนที่ต่อไป

booster - n. เครื่องยนต์เสริมของยานอวกาศที่ให้งำลังเพิ่มเติมสำหรับการเริ่มต้นการบิน

stage - n. ส่วนหนึ่งของจรวดที่บรรจุเครื่องยนต์และเชื้อเพลิง ซึ่งจะถูกปลดทิ้งเมื่อจรวดไปถึงระยะทางหนึ่งเหนือโลก

splash down - v. การที่ยานอวกาศกลับเข้าสู่ชั้นบรรยากาศโลกและลงจอดในแหล่งน้ำ

Reading Comprehension Quiz

1. What is the main focus of the article as indicated in the introductory paragraph?

- A. Predictions for space exploration trends in the upcoming year.
- B. A detailed history of space missions over the past five decades.
- C. A review of significant space exploration events from 2024.
- D. Upcoming challenges faced by major space agencies.

2. According to the article, which entity was responsible for the first U.S. moon landing in over 50 years?

- A. NASA
- B. JAXA
- C. Intuitive Machines
- D. The U.S. government directly

3. Which company developed the Odysseus spacecraft, and when did it permanently power down on the moon?

- A. NASA, in February
- B. Intuitive Machines, in late March
- C. JAXA, in January
- D. SpaceX, in October

4. What was the primary benefit of the data collected by Odysseus for NASA?

- A. To discover new forms of lunar life.
- B. To develop advanced communication systems for future missions.
- C. To better understand the moon's environment and improve landing techniques.
- D. To prepare for a human mission to the far side of the moon.

5. What initial problem did Japan's SLIM spacecraft encounter after its moon landing in January 2024?

- A. It failed to land in the correct location.
- B. It landed upside down, causing communication and power issues.
- C. It completely lost contact with Earth after touchdown.
- D. Its science experiments immediately malfunctioned.

6. How did JAXA determine the SLIM mission's precision landing experiment was successful?

- A. It was able to transmit data for several months.
- B. It discovered water ice near the landing site.
- C. It landed within 10 meters of its intended target.
- D. It managed to right itself after landing the wrong way up.

7. What unique achievement did China's Chang'e 6 spacecraft accomplish in late June?

- A. It was the first spacecraft to land on the far side of the moon.
- B. It launched the first private mission to the moon.
- C. It returned to Earth with rock and soil samples from the moon's far side.
- D. It successfully demonstrated new precision landing technology on the moon.

8. Why was the Chang'e 6 mission considered a significant achievement?

- A. It was the first mission to explore the surface of Jupiter's moon, Europa.
- B. It marked the first time any country collected and returned samples from the moon's far side.
- C. It deployed a new type of rover capable of exploring lunar caves.
- D. It used a completely reusable rocket system for its launch and return.

9. What is the primary destination and estimated travel time for the European Space Agency's Hera spacecraft, launched in October?

- A. Jupiter's moon Europa, a two-year journey.
- B. An asteroid system visited by NASA's DART, approximately two years.
- C. The International Space Station, a few weeks' journey.
- D. The far side of the moon, a six-month journey.

10. What is Hera's specific aim regarding the asteroid Dimorphos, following NASA's DART mission?

- A. To deflect Dimorphos from its current orbit.
- B. To collect samples from Dimorphos for Earth return.
- C. To closely examine how the DART strike affected the asteroid.
- D. To establish a permanent observation post on Dimorphos.

11. What is the main objective of NASA's Europa Clipper spacecraft, launched in October?

- A. To deliver supplies to the International Space Station.
- B. To test new asteroid deflection technologies.
- C. To search for possible signs of life on Jupiter's moon Europa.
- D. To map the surface of Mars in high resolution.

12. Why do scientists consider Jupiter's moon Europa a good candidate for supporting life?

- A. It has a thick atmosphere similar to Earth's.
- B. It is known to have active volcanoes on its surface.
- C. Studies suggest it likely contains a large ocean beneath its icy surface.
- D. It is located within the habitable zone of the solar system.

13. In early June, what was the primary purpose of Boeing's test mission for NASA?

- A. To launch a new satellite into Earth's orbit.
- B. To transport two American astronauts to the International Space Station.
- C. To test a new propulsion system in deep space.
- D. To demonstrate the capability of a reusable lunar lander.

14. What is the current plan for astronauts Butch Wilmore and Suni Williams, who were on the Boeing Starliner flight?

- A. They will return to Earth on the same Starliner spacecraft in August.
- B. They will remain on the ISS indefinitely due to technical issues.
- C. They are scheduled to return on a SpaceX Dragon spacecraft later.
- D. NASA has not yet decided on their return method or timeline.

15. What notable event occurred during one of SpaceX's Starship tests in October?

- A. The Starship's second stage failed to separate from the booster.
- B. The Super Heavy booster rocket was successfully "caught" by a ground structure.
- C. The Starship landed vertically on the moon for the first time.
- D. Astronauts were launched into orbit aboard the Starship system.